

Proof of the theorem that a metrizable space is compact if and only if it is complete in every metric¹⁾

V. Niemytzki and A. Tychonoff

Fréchet² calls a *metrizable* space R complete if it satisfies Cauchy's convergence criterion in at least one metric. Hausdorff³ has proved that R , considered with this metric, admits no *metric* extension. It is natural to ask under what conditions a metrizable space satisfies this completeness criterion in *every* metric compatible with its topological properties. In what follows, the answer stated in the title of this paper will be deduced from the general metrization criterion⁴ by a simple argument.

First, it is clear that every compact metrizable space is complete in every metric. Indeed, a compact metrizable space is known to be bicomact⁵ and therefore admits no extension, not only metrically but even topologically.⁶ Thus only the converse remains to be proved.

Let R be a space which is complete in some fixed metric, but is not compact. The theorem will follow if we show that R can always be remetrized so that metric completeness is lost.

Indeed, since R is not compact, it contains a countable divergent subset A consisting of points

$$(1) \qquad a_1, a_2, \dots, a_k, \dots,$$

On the other hand, since R is complete in the initially prescribed metric, we may assume, as is easily shown, that there is a fixed positive number $2d$ such that

$$\rho(a_i, a_k) \geq d$$

whenever i and k are distinct.

¹This paper answers a question posed by Mr. Alexandroff. We also owe to Mr. Alexandroff several simplifications of our original proof.

²See, for example, Fréchet, *Sur les ensembles abstraits*, Ann. Ec. Norm. 38 (1921), p. 341.

³Hausdorff, *Grundzüge der Mengenlehre* (Leipzig, 1914), p. 315. Hausdorff, *Mengenlehre* (Göschens Lehrbücherei, 1927), pp. 102 and 107.

⁴Alexandroff and Urysohn, *Une condition nécessaire et suffisante*, etc., Comptes Rendus Paris, 177 (1923), p. 1274.

⁵Alexandroff and Urysohn, *Zur Theorie der topologischen Räume*, Math. Ann., 92, p. 260.

⁶The work cited in footnote 5, pp. 261–263.

We now adjoin a new point ξ to the space R , regarded as a topological space, by specifying the countable neighborhood system

$$U_m(\xi) = \sum_{i=m}^{\infty} S\left(a_i, \frac{d}{4^m}\right) + \xi \quad (m = 1, 2, \dots, \text{indefinitely})$$

for this point.⁷ It is easy to verify that the resulting space $R + \xi$ is a topological space. Moreover, ξ is clearly a nonisolated point in R . To show that R is metrizable, it therefore suffices to exhibit a complete regular chain of covers in the sense of the Alexandroff–Urysohn metrization theorem.⁸ For this purpose, for each m we consider the cover Π_m of $R + \xi$ consisting of $U_m(\xi)$ and all the balls $S\left(x, \frac{d}{4^m}\right)$ (where x runs through all points of R and the metric used is that of the space R).

1) *The sequence of all Π_m is a complete chain of covers.*

Indeed, choose a member V_m of each family Π_m . Suppose that $V_1, V_2, \dots, V_m, \dots$ is decreasing and has nonempty intersection. If infinitely many of the V_m are $U_m(\xi)$, then their intersection can consist only of ξ , and the sequence in question is then identical to the system of all $U_m(\xi)$. If only finitely many of the V_m are $U_m(\xi)$, then, after the removal of finitely many terms, our sequence has the form

$$S\left(x_1, \frac{d}{4^n}\right), S\left(x_2, \frac{d}{4^{n+1}}\right), \dots, S\left(x_m, \frac{d}{4^{n+m}}\right), \dots$$

By assumption there is a point x belonging to all these balls. Thus the balls $S\left(x, \frac{d}{4^{n+m}}\right)$, ($m = 0, 1, 2, \dots$, indefinitely), form a full neighborhood system of x in R , and hence also in $R + \xi$.

2) *The sequence of all Π_m is regular.*

Indeed, from

$$S\left(x, \frac{d}{4^{m+1}}\right) \cdot S\left(y, \frac{d}{4^{m+1}}\right) \supset z,$$

it follows that

$$S\left(x, \frac{d}{4^{m+1}}\right) + S\left(y, \frac{d}{4^{m+1}}\right) \subset S\left(z, \frac{d}{4^m}\right)$$

and from

$$S\left(x, \frac{d}{4^{m+1}}\right) \cdot U_{m+1}(\xi) \neq 0,$$

it follows that for at least one $k \geq m + 1$,

$$S\left(x, \frac{d}{4^{m+1}}\right) \cdot S\left(a_k, \frac{d}{4^{m+1}}\right),$$

⁷The work cited in footnote 5, p. 261.

⁸The work cited in footnote 4, p. 1275.

and hence

$$S\left(x, \frac{d}{4^{m+1}}\right) + S\left(a_k, \frac{d}{4^{m+1}}\right) \subset S\left(a_k, \frac{3d}{4^{m+1}}\right) \subset U_m(\xi),$$

Since $U_{m+1}(\xi) \subset U_m(\xi)$ in any case, we also have

$$S\left(x, \frac{d}{4^{m+1}}\right) + U_{m+1}(\xi) \subset U_m(\xi),$$

This proves the regularity of the chain of covers $\{\Pi_m\}$.

The space $R + \xi$ can therefore be metrized. Its metric also induces a new metric on R , in which the sequence of points

$$a_1, a_2, \dots, a_k, \dots$$

is a divergent Cauchy sequence. Thus R is not complete in the new metric, as was to be proved.